WORKED SOLUTION

STEP III 2021, Question 9

Mechanics •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

The geometry is two applications of the sine rule, in the corner triangles CPQ and BRP; the shared height of QR above CB cancels when we multiply. For the bounces, a smooth wall keeps the velocity component along it and multiplies the component across it by e, so the angles the path makes with the wall before and after satisfy $\tan(\text{after}) = e\tan(\text{before})$. Each bounce then gives one of the two brackets in the first result.

Solution

The geometric identity

Let h be the distance of the line QR from BC. Because $QR \parallel CB$, triangle AQR is equilateral, so $CQ = BR = h/\sin 60^\circ = 2h/\sqrt{3}$.

In triangle CPQ the angle at C is 60° and the angle at P is θ, so the angle at Q is $120^\circ - \theta$. The sine rule gives

$$\frac{CP}{\sin(120^\circ - \theta)} = \frac{CQ}{\sin\theta} \quad \Longrightarrow \quad (a - x)\sin\theta = \frac{2h}{\sqrt{3}}\left(\tfrac{\sqrt{3}}{2}\cos\theta + \tfrac{1}{2}\sin\theta\right).$$

Dividing by $\sin\theta$ (non-zero, as $0 < \theta < 180^\circ$) and tidying,

$$\sqrt{3}\cot\theta + 1 = \frac{\sqrt{3}\,(a - x)}{h}.$$

In triangle BRP the angle at B is 60°, the angle at R is ϕ and the angle at P is $120^\circ - \phi$. The sine rule gives

$$\frac{BP}{\sin\phi} = \frac{BR}{\sin(120^\circ - \phi)} \quad \Longrightarrow \quad x\left(\tfrac{\sqrt{3}}{2}\cos\phi + \tfrac{1}{2}\sin\phi\right) = \frac{2h}{\sqrt{3}}\sin\phi,$$

so that

$$\sqrt{3}\cot\phi + 1 = \frac{4h}{\sqrt{3}\,x}.$$

Multiplying the two results, h and $\sqrt{3}$ cancel:

$$(\sqrt{3}\cot\phi + 1)(\sqrt{3}\cot\theta + 1) = \frac{4(a-x)}{x},$$

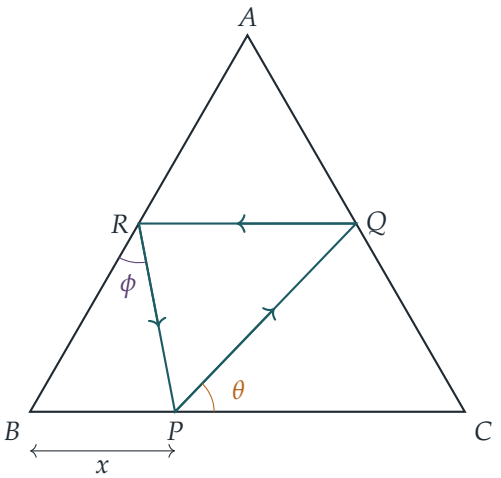
which is the required identity once we multiply by x.

The bounces

Model the ball as a particle moving on the smooth table, so it travels in straight lines between impacts, and the frame as smooth, so each impact gives only an impulse perpendicular to the wall. Suppose the ball meets a wall with speed u at angle α to it, and leaves with speed v at angle β to it. The component along the wall is unchanged and the component across it is reversed and multiplied by e:

$$v\cos\beta = u\cos\alpha, \qquad v\sin\beta = eu\sin\alpha \quad \Longrightarrow \quad \tan\beta = e\tan\alpha.$$

The ball also keeps moving the same way along the wall, so both α and β are acute when measured from that direction. (We need $e > 0$, otherwise the ball would slide along the wall after the first impact.)



At Q. The ball arrives along PQ, at angle $\angle PQC = 120^\circ - \theta$ to the wall CA, moving towards A. It leaves along QR, at angle $\angle AQR = 60^\circ$ to the wall (corresponding angles, as $QR \parallel CB$). So

$$\tan 60^\circ = e\tan(120^\circ - \theta).$$

Using the subtraction formula with $\tan 120^\circ = -\sqrt{3}$ and then multiplying top and bottom by $\cot\theta$,

$$\tan(120^\circ - \theta) = \frac{-\sqrt{3} - \tan\theta}{1 - \sqrt{3}\tan\theta} = \frac{\sqrt{3}\cot\theta + 1}{\sqrt{3} - \cot\theta}.$$

Write $c = \sqrt{3}\cot\theta$. The bounce condition $\sqrt{3}(\sqrt{3} - \cot\theta) = e(\sqrt{3}\cot\theta + 1)$ becomes $3 - c = e(c+1)$, that is, $4 - (c+1) = e(c+1)$, so

$$\sqrt{3}\cot\theta + 1 = \frac{4}{1+e}.$$

At R. The ball arrives along QR, at angle $\angle ARQ = 60^\circ$ to the wall AB, moving towards B, and leaves along RP at angle $\angle BRP = \phi$. So

$$\tan\phi = e\tan 60^\circ = \sqrt{3}\,e, \qquad \text{so} \qquad \sqrt{3}\cot\phi = \frac{1}{e}, \qquad \sqrt{3}\cot\phi + 1 = \frac{1+e}{e}.$$

Finding x. Substituting both brackets into the identity,

$$\frac{1+e}{e} \cdot \frac{4}{1+e} \cdot x = 4(a-x) \quad \Longrightarrow \quad x = e(a-x) \quad \Longrightarrow \quad x = \frac{ae}{1+e},$$

as required.

Going round again

Ask yourself What has to happen at P for the ball to set off along PQ a second time?

The ball reaches P along RP. In triangle BRP this path makes angle $\angle RPB = 120^\circ - \phi$ with BC, so it arrives moving towards C only if this angle is acute. To retrace PQ, it must leave at angle θ to PC, so we need

$$\tan\theta = e\tan(120^\circ - \phi).$$

From the work at Q, $\sqrt{3}\cot\theta = \frac{4}{1+e} - 1 = \frac{3-e}{1+e}$, so $\tan\theta = \dfrac{\sqrt{3}(1+e)}{3-e}$, which is positive: PQ does head towards the C side.

The same formula as at Q, with ϕ in place of θ and $\sqrt{3}\cot\phi = \frac{1}{e}$, gives

$$\tan(120^\circ - \phi) = \frac{\sqrt{3}\,(\sqrt{3}\cot\phi + 1)}{3 - \sqrt{3}\cot\phi} = \frac{\sqrt{3}\,(1+e)/e}{(3e-1)/e} = \frac{\sqrt{3}(1+e)}{3e-1}.$$

If $e \le \frac{1}{3}$, then $120^\circ - \phi \ge 90^\circ$, so after the impact the ball moves straight away from BC or back towards B. It cannot then follow PQ, which leans towards C. So $e > \frac{1}{3}$, and the condition becomes

$$\frac{\sqrt{3}(1+e)}{3-e} = \frac{\sqrt{3}\,e(1+e)}{3e-1} \quad \Longleftrightarrow \quad 3e - 1 = 3e - e^2 \quad \Longleftrightarrow \quad e^2 = 1.$$

Since $e > 0$, this gives $e = 1$, as required. The ball's speed changes at each impact (unless $e = 1$), but speed plays no part here: only the directions matter.

Checks

- When $e = 1$: $x = \frac{a}{2}$ and $\tan\theta = \tan\phi = \sqrt{3}$, so $\theta = \phi = 60^\circ$ and PQR is the triangle joining the midpoints of the sides. By symmetry it is a closed perfectly elastic orbit, as expected.
- With $a = 1$ and $e = \frac{1}{2}$: $x = \frac{1}{3}$ and $\theta \approx 46.10^\circ$. Stepping the ball through the three impacts by computer brings it back to P exactly. After the next impact it leaves at about 68.95° to BC, not 46.10°, so it does not go round PQR again.
- As $e \to 0$, $x \to 0$: P slides into the corner B. This fits the ball losing almost all its velocity across each wall.

Common slips

- Applying e to the whole velocity, or to the component along the wall. Only the component perpendicular to the smooth wall changes.
- Measuring the angle of incidence from the normal and then using $\tan\beta = e\tan\alpha$, which holds for angles measured from the wall. Measured from the normal, the law is $\cot\beta = e\cot\alpha$.
- In the final part, forgetting that $120^\circ - \phi$ might be obtuse, or that $e = -1$ must be rejected.

Beyond STEP

Paths of a ball bouncing inside a region are the subject of *mathematical billiards*. With $e = 1$ the triangle joining the midpoints is the Fagnano orbit. In any acute triangle this orbit joins the feet of the altitudes, and it is the inscribed triangle of shortest perimeter. With $e < 1$ each bounce flattens the path towards the wall, and as we found here, the ball cannot repeat a closed circuit. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats impulse and collisions carefully. For billiards themselves, Tabachnikov's *Geometry and Billiards* is a lovely next step.

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